

Preservation and Represervation Specifications

Engines and engine-generator sets

- **Diesel engines – all MTU series**
- **Gas engines – MTU Series 4000**
- **Diesel engine-generator sets with MTU engines of Series 1600, 2000, 4000**
- **Gas engine-generator sets with MTU engines of Series 400 and 4000**

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Power. Passion. Partnership.

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1 General Information

1.1 Notes on the use and validity of the Preservation and Re-preservation Specifications

Used symbols and means of representation

The following instructions are highlighted in the text and must be observed:

Important

This field contains product information which is important or useful for the user. It refers to instructions, work and activities that have to be observed to prevent damage or destruction to the material.

Note:

A note provides special instructions that must be observed when performing a task.

Applicability of this publication

The current amendment status of these Preservation and Re-preservation Specifications is shown in the Revision overview (→ Page 106).

The Preservation and Re-preservation Specifications are modified or supplemented as required. Before using them, make sure you have the latest version. Prior to using these specifications, check with the customer service of MTU Friedrichshafen GmbH or MTU Onsite Energy to ensure that you have the latest version of this publication (publication number A001070/..).

If you have further queries, please contact your MTU representative.

Other applicable documents

- Preservation specifications of the gearbox manufacturer
- Documentation of the engine or engine-generator set
- Safety data sheets for preservation media
- Current MTU Fluids and Lubricants Specifications

Validity of the preservation specifications

These preservation specifications define the media for preservation/re-preservation (preservatives) and the guidelines for de-preservation and packaging of engines and engine-generator sets from MTU.

The preservation specifications apply to the following products:

- All diesel and gas engines from MTU Friedrichshafen GmbH
 - Reserve stock engines
 - Installed engines that have not yet been put into service
 - Field engines with extended downtime, e.g. normal out-of-service periods, out-of-service-periods for scheduled maintenance or unscheduled repair work.
- Diesel engine-generator sets with MTU engines of Series 1600, 2000, 4000 from MTU Onsite Energy
 - Engine-generator sets stored in reserve
 - Installed engine-generator sets that have not yet been put into service
 - Engine-generator sets in the field with long downtimes, e.g. normal out-of-service periods, out-of-service-periods for scheduled maintenance or unscheduled repair work
- Gas engine-generator sets with MTU engines of Series 400 and 4000 from MTU Onsite Energy
 - Engine-generator sets stored in reserve
 - Installed engine-generator sets that have not yet been put into service
 - Engine-generator sets in the field with long downtimes, e.g. normal out-of-service periods, out-of-service-periods for scheduled maintenance or unscheduled repair work

The preservation / re-preservation scope is the same for all engine models and engine-generator sets.

When preserving complete gensets, the preservation specifications of component manufacturers, e.g. of the gearbox, generator, preheating unit and couplings, must also be observed.

In addition to these preservation specifications, the corresponding documentation of the engine or engine-generator set must also be observed. Work and tests during an interruption of operation and prior to a renewed startup are to be performed according to this documentation.

The following described measures are limited to engines and engine-generator sets from MTU

Exception: For the 1800 series, they refer to the complete PowerPack®.

Warranty

Use of the approved preservatives, either under the brand name or in accordance with the specifications given in this publication, constitutes part of the warranty conditions.

The supplier of the preservatives is responsible for the worldwide standard quality of the named products.

MTU Friedrichshafen GmbH and MTU Onsite Energy accept no liability whatsoever for improper or illegal use of the preservatives which they have approved. Users of the products named in these specifications are therefore obliged to inform themselves of the locally valid regulations.

Important

As the re-preservation intervals and the medium also depend on the storage conditions and type of packaging, the warranty shall become invalid in the event of incorrect storage or packaging.

These regulations are derived from the manufacturers' specifications, statutory regulations and technical guidelines valid in the individual countries. Great differences can apply from country to country and a generally valid guide to applicable regulations for preservatives is therefore not possible within this publication.

Users of the products named in these Fluids and Lubricants Specifications are therefore obliged to inform themselves of the locally valid regulations. MTU accepts no liability whatsoever for improper or illegal use of the preservatives which it has approved.

Registered trademarks

All brand names are registered trademarks of the manufacturer concerned.

1.2 Safety instructions for handling preservatives

Important

Preservatives for drive systems can be inherently dangerous. Certain regulations must be observed when handling, storing and disposing of these substances.

These regulations are contained in the manufacturers' instructions, legal requirements and technical guidelines valid in the individual countries. Great differences can apply from country to country and a generally valid guide to applicable regulations for preservatives is therefore not possible within this publication.

Users of the products named in these specifications are therefore obliged to inform themselves of the locally valid regulations. MTU Friedrichshafen GmbH or MTU Onsite Energy accept no responsibility whatsoever for improper or illegal use of the preservatives which it has approved.

2 Preservation and Represervation

2.1 Notes on preservation and re-preservation

Preservation

The term "preservation" refers to:

- Initial preservation in the plant
- Preservation of an engine or engine-generator set in the field prior to long out-of-service periods

Prior to delivery from the plant, MTU engines are usually initially preserved in accordance with MTV 5073 (does not apply to Series 400 engines). This is generally carried out during the acceptance test on the test bench.

In event of extended out-of-service periods, machined, non-protected surfaces, for example cylinder liner running faces, are susceptible to corrosion and must therefore be preserved.

Preservation measures are carried out in the same manner as re-preservation. The scope of preservation depends on the duration of the out-of-service period.

Important

Engine-generator sets, field engines and engines that were put out of service for a scheduled major overhaul must be preserved immediately after their last service period.

Important

Coolant circuits must always be preserved after the coolant has been drained. Preservation is not required if the coolant is left in the system.

Preservation must be repeated at regular intervals (re-preservation).

Re-preservation

"Re-preservation" refers to the renewal of already existing preservation at specified time intervals.

The intervals differ for internal and external preservation and – for internal preservation – according to the different media (oil, fuel, coolant) and filling levels. For re-preservation intervals, refer to (→ Page 25) and (→ Page 26)

Important

As the re-preservation intervals and the medium also depend on the storage conditions and type of packaging, the warranty shall become invalid in the event of incorrect storage or packaging.

For storage and transport conditions, see (→ Page 22).

Preservatives

The service life, operational reliability and function of the drive systems are largely dependent on the fluids, lubricants and preservatives employed. The correct selection and treatment of these fluids, lubricants and preservatives are therefore extremely important.

For preservation and re-preservation of MTU engines, only the preservatives approved in these preservation specifications must be used (→ Page 13).

Documentation requirements

(Re-)preservation, de-preservation and monitoring of an engine with climate-compatible packaging must be documented by the person responsible:

- Check sheet for preservation and represervation of engines (→ Page 10)
- Check sheet for preservation and re-preservation of engine-generator sets (→ Page 11)
- Instruction sheet for de-preservation if climate-compatible packaging is used (→ Page 82)
- Monitoring Sheet for engines with climate-compatible packaging (→ Page 94)

2.2 Check sheet for preservation and represervation of engines

Important

All preservation work that is performed must be recorded on the check sheet shown below. For engines stored with a manufacturer guarantee, the Monitoring Sheet (→ Page 94) must also be filled out and sent back to MTU Friedrichshafen GmbH in due time before putting the engine back into operation.

Correct completion of the preservation tasks as specified in the preservation specifications must be certified on this check sheet by the person completing the various tasks.

Engine model:	Engine No.:	Acceptance date:
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Tasks performed	Date	Name
Lube oil system - Preserved with preservation oil - Oil brand used:		
Fuel system - Preserved with preservative fuel - Fuel filter, prefilter and fuel lines have not been drained - Preservative fuel used:		
Cooling system, filled - Preserved with treated coolant - Coolant not drained - Brand of coolant used:		
Cooling system, unfilled - Preserved with an approved corrosion inhibitor for internal preservation of the coolant circuit - Corrosion inhibitor drained - Corrosion inhibitor used:		
Combustion chamber - Preserved with preservation oil - Oil brand used:		
Non-painted parts - Painted with corrosion inhibitor - Engine openings sealed as specified - All parts to be kept free of paint, such as flywheel, starter ring gear and starter pinion as well as non-painted sections of the control linkage and the uncovered coupling flange (as appropriate) for the three-phase generator are brush-coated with corrosion-inhibiting oil. - Corrosion inhibitor used:		
Engine is preserved as specified.		
Re-preservation performed according to instructions.		

Table 1:

Further procedure

1. After completing preservation work, put the completed check sheet into a closed plastic sleeve.
2. Seal the sleeve at the open side and attach it to the engine so that it is clearly visible.
3. Store the check sheet until de-preservation beside the engine and send back to MTU Friedrichshafen GmbH in good time before initial operation.